Complex Analysis

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The motivation for complex numbers is quite established as the field extension $\mathbb{R} \subset \mathbb{R}[i] \simeq \mathbb{C}$ which gives algebraic closure.

1 The Complex Numbers

The next field that will be particularly important is the complex numbers. It is straightforward to construct $\mathbb{C}$, but let's motivate this for a minute.

Example 1.1 (Polynomial Roots)

The roots of the polynomial

$$f(x) = x^2 + 1 \tag{1}$$

does not exist in $\mathbb{R}$.

Therefore, we would like to construct a new space that contains all possible roots for all possible polynomials with real coefficients. We call this $\mathbb{C}$. Clearly, by constructing polynomials of the form $x^2 - r^2$ for some $r \in \mathbb{R}$, we know that $\mathbb{R} \subset \mathbb{C}$. Therefore, we want to create a further extension of $\mathbb{R}$, along with some canonical injection $\iota : \mathbb{R} \to \mathbb{C}$ that is also a field homomorphism. It turns out that once we construct this field, there is no possible way that we can make it an ordered field. However, the norm extends naturally into $\mathbb{C}$ such that ι is isometric. Finally, we can define a new operator called *conjugation* that gives us additional structure.

This is not the only way to construct the complex plane however. Rather than defining all these from scratch, we could just define the addition operations with an isometric vector space isomorphism from $\mathbb{R}^2$ to $\mathbb{C}$ actually, and then define multiplication. Another way is to start again with $\mathbb{Q} \times \mathbb{Q}$, define a norm on it, complete it, and finally define the addition and multiplication operations that satisfy the field property.

1.1 Construction

Theorem 1.1 (Construction of the Complex Numbers)

Let $\mathbb{C}$ be defined as the space $\mathbb{R} \times \mathbb{R}$ with the following operations.

1. *Addition.* $x = (a, b), y = (c, d) \implies x +_{\mathbb{C}} y = (a + c, b + d)$.
2. *Additive Identity.* $0_{\mathbb{C}} = (0, 0)$.
3. *Additive Inverse.* $x = (a, b) \implies -x = (-a, -b)$.
4. *Multiplication.* $x = (a, b), y = (c, d) \implies x \times_{\mathbb{C}} y = (ac - bd, ad + bc)$.
5. *Multiplicative Identity.* $1_{\mathbb{C}} = (1, 0)$.
6. *Multiplicative Inverse.*

$$x = (a, b) \implies x^{-1} = \left(\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right) \tag{2}$$

Our first claim is that $(\mathbb{C}, +_{\mathbb{C}}, \times_{\mathbb{C}})$ is a field. Furthermore, we define the additional structures

1. *Conjugate.* $x = (a, b) \implies \overline{x} = (a, -b)$.
2. *Norm.* $|x|_{\mathbb{C}} = x \times_{\mathbb{C}} \overline{x} = a^2 + b^2$.
3. *Metric.* This is the norm-induced metric. $d_{\mathbb{C}}(x, y) = |x - y|_{\mathbb{C}}$.
4. *Topology.* This is the metric-induced topology generated by the open balls $B(x, r) := \{y \in \mathbb{C} | d(x, y) < r\}$, where $x \in \mathbb{C}, r \in \mathbb{R}$.

Our second claim is that the canonical injection $\iota : \mathbb{R} \to \mathbb{C}$ defined

$$\iota(r) = (r, 0) \tag{3}$$

is an isometric field isomorphism. Our third claim is that $\mathbb{C}$ is Cauchy-complete with respect to this metric.

Note that we do not talk about order $\mathbb{C}$, and so the concepts of Dedekind completeness, least upper bound properties, or Archimedean principle is meaningless in the complex plane.

Definition 1.1 (Imaginary Number)

Let us denote $i = (0, 1)$ which we call the **imaginary number**, which has the property that $i^2 = 1$. With this notation, we can see through abuse of notation that

$$(a, b) = (a, 0) + (0, b) = (a, 0) + (b, 0)(0, 1) = a + bi \tag{4}$$

Therefore, we generally write complex numbers as $z = a + bi$, and we define the real and imaginary components as $\text{Re}(z)$ and $\text{Im}(z)$, respectively.

Note that the identity $x^2 + 1 \equiv (x + i)(x - i)$ implies that the equation $x^2 = -1$ has exactly two solutions in $\mathbb{C}$, i and $-i$. Therefore, if a subfield of $\mathbb{C}$ contains one of these solutions, it must contain the other (since i and $-i$ are additive and multiplicative inverses).

Furthermore, since i is defined to be $\sqrt{-1}$, we could replace i with $-i$ and our calculations would still be consistent throughout the rest of mathematics. In fact, i and $-i$ behave **exactly** identically and cannot be distinguished in an abstract sense. Visually, the complex plane "flipped" across the real number axis produces the same complex plane.

Theorem 1.2 (Uniqueness of $\mathbb{C}$)

$\mathbb{C}$ is unique up to an isomorphism that maps all real numbers to themselves. Every complex number can be uniquely written as $a + bi$, where $a, b \in \mathbb{R}$ and i is a fixed element such that $i^2 = -1$.

Proof. Consider the subset of $\mathbb{C}$

$$K \equiv \{a + bi \mid a, b \in \mathbb{R}\} \tag{5}$$

By evaluating its operations, we can check for closure, identity, and invertibility of nonzero elements to conclude that K is a subfield of $\mathbb{C} \implies$ by prop. (iii), $K = \mathbb{C} \implies$ every element in $\mathbb{C}$ can be written in form $a + bi$. To prove uniqueness, we assume that $p \in \mathbb{C}$ can be written in distinct forms $p = a + bi = a' + b'i$. Then

$$\begin{aligned} a + bi = a' + b'i &\implies (a - a')^2 = (b'i - bi)^2 = -(b' - b)^2 \\ &\implies a - a' = b' - b = 0 \end{aligned}$$

To prove uniqueness of $\mathbb{C}$ up to ismorphism, we assume that $\mathbb{C}'$ exists with i' such that i'^2 containing elements $a + bi'$. Let $f : \mathbb{C} \to \mathbb{C}'$ defined

$$f(a + bi) = a + bi' \tag{6}$$

Then,

$$\begin{aligned} f\big((a + bi) + (c + di)\big) &= f\big((a + c) + (b + d)i\big) \\ &= (a + c) + (b + d)i' \\ &= (a + bi') + (c + di') \\ &= f(a + bi) + f(c + di) \\ f\big(\kappa(a + bi)\big) &= f\big(\kappa a + \kappa bi\big) \\ &= \kappa a + \kappa bi' \\ &= \kappa(a + bi') \\ &= \kappa f(a + bi) \end{aligned}$$

So, f is an isomorphism, and $\mathbb{C} \simeq \mathbb{C}'$. From analysis, we can construct and prove the existence of $\mathbb{R}$. We then define the map

$$\rho : \mathbb{R}^2 \to \mathbb{C},\ \rho(a, b) \equiv a + bi \tag{7}$$

with $\rho(1,0)$ as the multiplicative identity and $\rho(0,1) \equiv i$. Therefore, every element of $\mathbb{C}$ can be uniquely represented as an element of $\mathbb{R}^2$.

Unfortunately, we lose the ordering.

Theorem 1.3 (Order on Complex Plane)

There exists no order on $\mathbb{C}$ that makes it a totally ordered field.

Proof. We attempt to construct an order on i and 0 in $\mathbb{C}$.
1. If $i = 0$, then $i^4 = 0 \cdot i^3 \implies 1 = 0$, which contradicts that $0 < 1$.
2. If $i \neq 0$, then $i^2 > 0$ from the field axioms, and so $-1 > 0$. But this also means that $1 = i^4 > 0$. This contradicts the ordered field property that $x > 0 \iff -x < 0$.

Therefore $\mathbb{C}$ cannot be turned into an ordered field.

1.2 Properties of the Complex Plane

Theorem 1.4 (Conjugation is an Isomorphism)

Conjugation is an isometric field automorphism of $\mathbb{C}$.

$$c = a + bi \mapsto \bar{c} = a - bi \tag{8}$$

This is identically defined by replacing i with $-i$. Clearly, $\bar{\bar{c}} = c$.

Proof.

Theorem 1.5 (Properties of Conjugation)

For any $c \in \mathbb{C}$, $c + \bar{c}$ and $c\bar{c}$ are real.

Proof. Using the fact that the complex conjugate is an isomorphism,

$$c + \bar{c} = \bar{c} + \bar{\bar{c}} = \bar{c} + c = c + \bar{c}$$
$$c\bar{c} = \bar{c}\bar{\bar{c}} = \bar{c}c = c\bar{c}$$

Note that we proved this abstractly using only the properties given above, and did not decompose c to its **algebraic form** $a + bi$.

If $c = a + bi,\ a, b \in \mathbb{R}$, then

$$c + \bar{c} = 2a,\ c\bar{c} = a^2 + b^2 \tag{9}$$

1.3 Polar Coordinates

In case the reader is unaware, it is common to interpret complex numbers $c = a + bi$ as points or vectors (a, b) on the complex plane.

Definition 1.2 (Polar Form of Complex Numbers)

The **polar representation**, or **trigonometric representation**, of a complex number $c = a + bi$ is defined using the equations

$$a = r\cos\varphi,\ b = r\sin\varphi \implies c = r(\cos\varphi + i\sin\varphi) \tag{10}$$

where $r = |c|$ and φ is the **argument** of c, which is the angle formed by the corresponding vector with the polar axis defined within the interval $[0, 2\pi)$.

$$\arg(c) \equiv \tan^{-1}\frac{b}{a} \tag{11}$$

This mapping can be defined

$$\rho : \mathbb{R} \times \frac{\mathbb{R}}{2\pi} \to \mathbb{C},\ \rho(r, \varphi) = r(\cos\varphi + i\sin\varphi) \tag{12}$$

Theorem 1.6 (The Argument as a Quasi-Homomorphism)

ρ is "similar" to a homomorphism in the following way. By defining the domain and codomain as groups,

$$\rho : \big(\mathbb{R}, \times\big) \times \Big(\frac{\mathbb{R}}{2\pi}\Big) \to \big(\mathbb{C}, \times\big) \tag{13}$$

we can see that

$$\rho(r_1, \varphi_1) \times \rho(r_2, \varphi_2) = \rho(r_1 \times r_2, \varphi_1 + \varphi_2) \tag{14}$$

or equivalently,

$$r_1(\cos\varphi_1 + i\sin\varphi_1) \cdot r_2(\cos\varphi_2 + i\sin\varphi_2) = r_1r_2(\cos(\varphi_1 + \varphi_2) + i\sin(\varphi_1 + \varphi_2)) \tag{15}$$

Corollary 1.7 (Ratio of Complex Numbers in Polar Form)

The formula for the ratio of complex numbers is defined

$$\frac{r_1(\cos\varphi_1 + i\sin\varphi_1)}{r_2(\cos\varphi_2 + i\sin\varphi_2)} = \frac{r_1}{r_2}(\cos(\varphi_1 - \varphi_2) + i\sin(\varphi_1 - \varphi_2)) \tag{16}$$

Corollary 1.8 (De Moivre's Formula)

The positive integer power of a complex number can be written using **De Moivre's formula**.

$$\big(r(\cos\varphi + i\sin\varphi)\big)^n = r^n(\cos n\varphi + i\sin n\varphi) \tag{17}$$

1.4 Roots, Exponentials, Logarithms

We can use this formula to extract a root of nth degree from a complex number $c = r(\cos\varphi + i\sin\varphi)$, which means to solve the equation $z^n = c$. Let $z = s(\cos\psi + i\sin\psi)$. Then by De Moivre's formula,

$$\begin{aligned} z^n &= s^n(\cos n\psi + i\sin n\psi) = r(\cos\varphi + i\sin\varphi) \\ &\implies s = \sqrt[n]{r},\ \psi = \frac{\varphi + 2\pi k}{n} \\ &\implies z = \sqrt[n]{r}\left(\cos\frac{\varphi + 2\pi k}{n} + i\sin\frac{\varphi + 2\pi k}{n}\right) \text{ for } k = 0, 1, ..., n-1 \end{aligned}$$

Geometrically, the n solutions lie at the vertices of a regular n-gon centered at the origin. When $c = 1$, the solutions are the nth roots of unity.

1.5 Trigonometric Functions

Now with complex numbers, we have a yet another way of defining trigonometric functions that generalizes that of the reals. We can use the series representation.

1.6 Dual Numbers

Another similar number system.

2 TBD

2.1 Algebraic Extension of Field $\mathbb{R}$

We introduce the number i, called the **imaginary unit**, such that $i^2 = -1$. We may multiply real numbers y to i to get yi, and we can add real numbers to such numbers, to get numbers of the form

$$x + yi, \quad x, y \in \mathbb{R} \tag{18}$$

We then define all objects of the form $x + iy$ as the **complex numbers**, with addition defined

$$(x_1 + iy_1) + (x_2 + iy_2) \equiv (x_1 + x_2) + i(y_1 + y_2) \tag{19}$$

and multiplication defined

$$(x_1 + iy_1) \cdot (x_2 + iy_2) \equiv (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1) \tag{20}$$

As expected, this makes $+$ and $\cdot$ commutative operations. Furthermore, two complex numbers $z = x_1 + iy_1$ and $w = x_2 + iy_2$ are equal if and only if $x_1 = x_2$ and $y_1 = y_2$.

One nontrivial property of field $\mathbb{C}$ is that every element $z \in \mathbb{C}$ has a multiplicative inverse z^{-1}. To find this, we must define the following.

Definition 2.1 (Complex Conjugate)

Given complex number $z = x + iy$, its **complex conjugate** is

$$\overline{z} = \overline{x + iy} = x - iy \tag{21}$$

Note that

$$z \cdot \overline{z} = x^2 + y^2 \neq 0 \text{ iff } z \neq 0 \tag{22}$$

Thus, given z,

$$z^{-1} = \frac{1}{z \cdot \overline{z}} \cdot \overline{z} \iff (x + yi)^{-1} = \frac{x}{x^2 + y^2} - i\frac{y}{x^2 + y^2} \tag{23}$$

2.1.1 Geometric Interpretation of $\mathbb{C}$

Once the algebraic operations $+$ and $\cdot$ has been introduced, the symbol i is no longer needed. That is, we can define a new set $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ with the operations $+_{\mathbb{R}}, \cdot_{\mathbb{R}} : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^2$ defined

$$\begin{aligned}(x_1, y_1) +_{\mathbb{R}} (x_2, y_2) &\equiv (x_1 + x_2, y_1 + y_2) \\ (x_1, y_1) \cdot_{\mathbb{R}} (x_2, y_2) &\equiv (x_1x_2 - y_1y_2, x_1y_2 + x_2y_1)\end{aligned}$$

We can check that this new set $(\mathbb{R}^2, +_{\mathbb{R}}, \cdot_{\mathbb{R}})$ is isomorphic to $(\mathbb{C}, +, \cdot)$ as fields, and therefore one can identify complex numbers with vectors $z = (x, y)$ of the plane $\mathbb{R}^2$, where $x = \operatorname{Re} z$ is called the **real part** and $y = \operatorname{Im} z$ is called the **imaginary part**.

Definition 2.2 (Norm, Metric of $\mathbb{C}$)

Moreover, the isomorphism

$$\gamma : \mathbb{C} \to \mathbb{R}^2, \;\; \gamma(x + yi) = (x, y) \tag{24}$$

induces additional structures on $\mathbb{C}$, such as the norm and metric.

1. The norm of $z = x + iy \in \mathbb{C}$ is defined as the norm of $\gamma(z) = (x, y) \in \mathbb{R}^2$. That is,

$$|z| = |x + yi| = |(x, y)| = \sqrt{x^2 + y^2} \tag{25}$$

Or more simply,

$$|z| = z \cdot \overline{z} \tag{26}$$

2. The metric of two complex numbers $z_1, z_2 \in \mathbb{C}$ is defined

$$|z_1 - z_2| = |(x_1, y_1) - (x_2, y_2)| = |(x_1 - x_2, y_1 - y_2)| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \tag{27}$$

Or more simply,

$$|z_1 - z_2| = (z_1 - z_2) \cdot \overline{(z_1 - z_2)} \tag{28}$$

Definition 2.3 (Polar Coordinates of $\mathbb{C}$)

Given the basis transformation of polar coordinates $(r, \varphi) \mapsto p(r, \varphi) = (x, y)$ where

$$p\begin{pmatrix} r \\ \varphi \end{pmatrix} = \begin{pmatrix} r\cos\varphi \\ r\sin\varphi \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \tag{29}$$

the isomorphism $\mathbb{C} \simeq \mathbb{R}^2$ induces a similar polar transformation in $\mathbb{C}$

$$\rho = \gamma^{-1} \circ p \circ \gamma : \mathbb{C}_{(r,\theta)} \to \mathbb{C}_{(x,y)}, \quad \rho(r + \theta i) = r\cos\theta + r\sin\theta i = x + yi \tag{30}$$

as shown in the commutative diagram.

$$\begin{array}{ccc} \mathbb{C}_{(r,\theta)} & \xrightarrow{\rho} & \mathbb{C}_{(x,y)} \\ \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \gamma} \\ \mathbb{R}^2_{(r,\theta)} & \xrightarrow{p} & \mathbb{R}^2_{(x,y)} \end{array} \tag{31}$$

Therefore, we can write

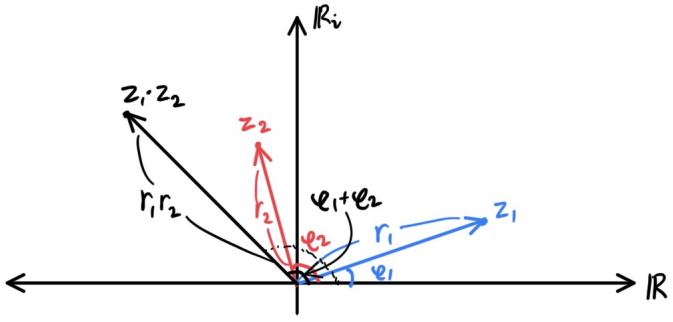
$$z = r(\cos\varphi + i\sin\varphi) \tag{32}$$

where $r = |z|$ is called the **magnitude** of z, and $\varphi = \operatorname{Arg} z$ is called the **argument** of z.

Lemma 2.1 (Multiplication of Complex Numbers in Polar Form)

It turns out that multiplication is a lot easier in polar coordinates than in rectangular ones:

$$\begin{aligned} z_1 \cdot z_2 &= (r_1\cos\varphi_1 + ir_1\sin\varphi_1)(r_2\cos\varphi_2 + ir_2\sin\varphi_2) \\ &= \dots \\ &= r_1 r_2\big(\cos(\varphi_1 + \varphi_2) + i\sin(\varphi_1 + \varphi_2)\big) \end{aligned}$$



Theorem 2.2 (De Moivre's Formula)

By induction using the previous lemma, we get

$$z = r(\cos\varphi + i\sin\varphi) \implies z^n = r^n(\cos n\varphi + i\sin n\varphi) \tag{33}$$

Corollary 2.3 (Roots of Unity)

The n complex solutions of the equation

$$z^n = a \tag{34}$$

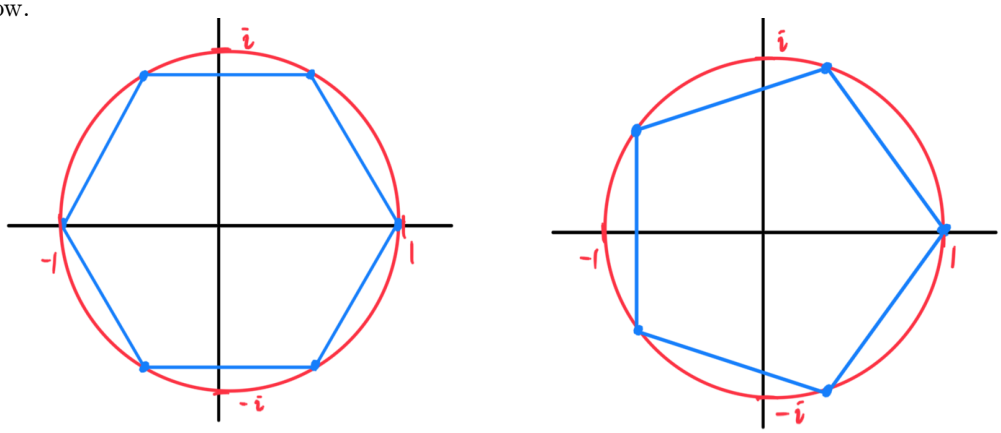
where $a = \rho(\cos\psi + i\sin\psi)$ is

$$z_k = \sqrt[n]{\rho}\left(\cos\left(\frac{\psi + 2\pi k}{n}\right) + i\sin\left(\frac{\psi + 2\pi k}{n}\right)\right), \qquad k = 0, 1, 2, \ldots, n-1 \tag{35}$$

Moreover, if $a = 1$, then the n complex solutions are called the n**th roots of unity**, defined

$$z_k = \cos\left(\frac{2\pi k}{n}\right) + i\sin\left(\frac{2\pi k}{n}\right), \qquad k = 0, 1, 2, \ldots, n-1 \tag{36}$$

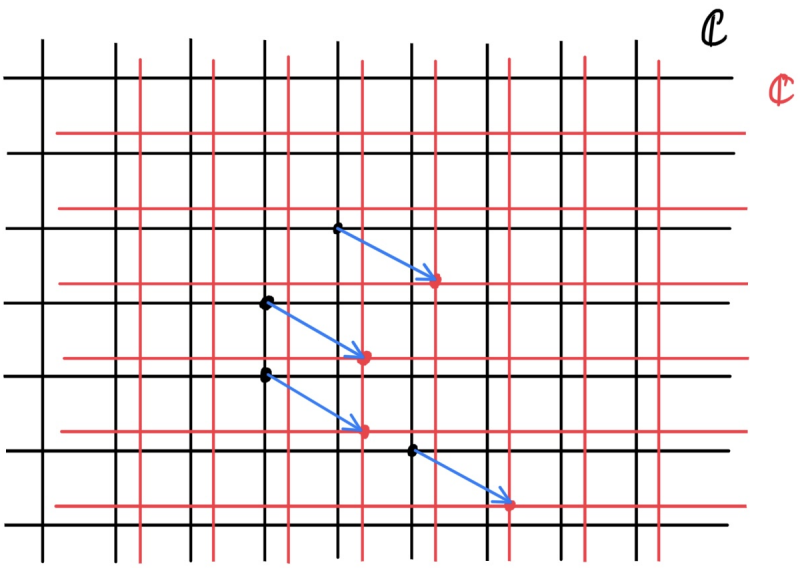
which shows that the nth roots of unity are at the vertices of a regular n-sided polygon inscribed in the unit circle, with one vertex at 1, within the complex plane. The 5th and 6th roots of unity are shown below.



Finally, we can visualize certain transformations in $\mathbb{C}$. For a fixed $b \in \mathbb{C}$, the sum $z + b$ cam be interpreted as the mapping of $\mathbb{C}$ onto itself given by the formula

$$z \mapsto z + b \tag{37}$$

This mapping is a translation of the plane by the vector b.



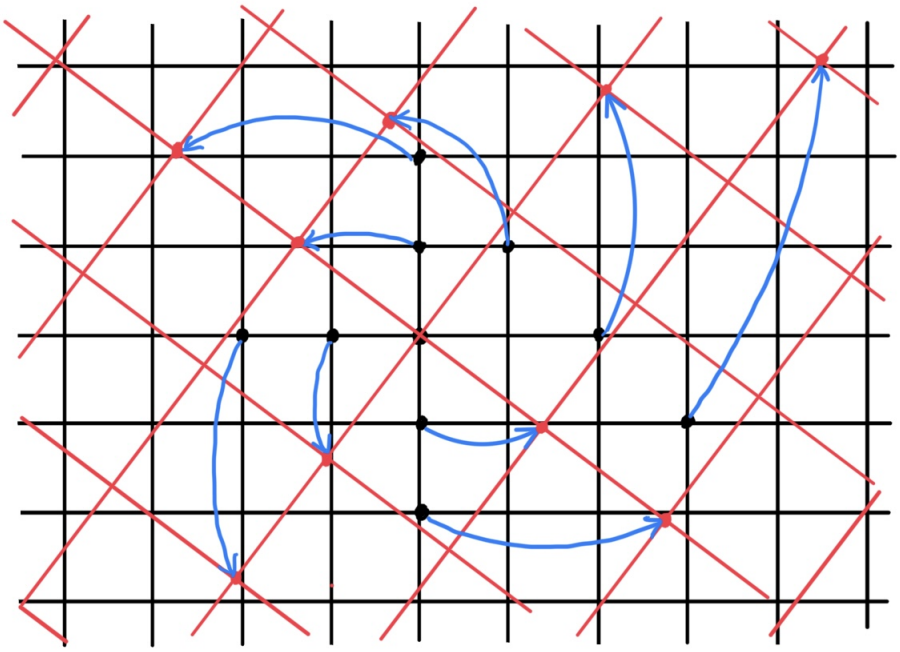
Visualizing multiplication is a bit harder. Given a

$$a = |a|(\cos\varphi + i\sin\varphi) \neq 0 \tag{38}$$

the product az can be interpreted as the mapping of $\mathbb{C}$ onto itself given by the formula

$$z \mapsto az \tag{39}$$

which is the composition of a dilation by a factor of $|a|$ and a rotation through the angle $\varphi \in \operatorname{Arg} a$.

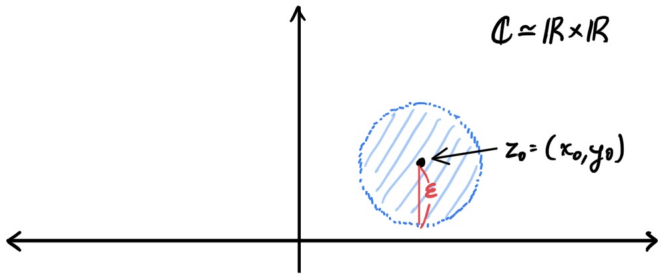


2.2 Sequences and Series in $\mathbb{C}$

Our previous construction of a metric within $\mathbb{C}$ enables to define the ϵ-neighborhood of a number $z_0 \in \mathbb{C}$ as the set

$$U_\epsilon(z_0) \equiv \{z \in \mathbb{C} \mid |z - z_0| < \epsilon\} \tag{40}$$

which can be visualized as an open disk of radius ϵ in $\mathbb{R}^2$ centered at point (x_0, y_0) if $z_0 = x_0 + iy_0$.



Definition 2.4 (Convergence of a Sequence in $\mathbb{C}$)

A sequence $\{z_n\}$ of complex numbers **converges** to $z_0 \in \mathbb{C}$ if and only if

$$\lim_{n\to\infty} |z_n - z_0| = 0 \tag{41}$$

It is clear from the inequality

$$\max\{|x_n - x_0|, |y_n - y_0|\} \leq |z_n - z_0| \leq |x_n - x_0| + |y_n - y_0| \tag{42}$$

that a sequence of complex numbers converges if and only if the sequences of its real and imaginary parts of the terms of the sequence both converge. That is,

$$\{z_n\} \text{ converges} \iff \{\operatorname{Re} z\} \text{ and } \{\operatorname{Im} z\} \text{ converges} \tag{43}$$

Lemma 2.4 (Convergence of Cauchy Sequences over $\mathbb{C}$)

A sequence of complex numbers $\{z_n\}$ is called a **Cauchy sequence** if for every $\epsilon > 0$ there exists an index $N \in \mathbb{N}$ such that

$$|z_n - z_m| < \epsilon \text{ for all } n, m > N \tag{44}$$

It is also clear that

$$\{z_n\} \text{ is Cauchy} \iff \{\operatorname{Re} z\} \text{ and } \{\operatorname{Im} z\} \text{ is Cauchy} \tag{45}$$

and using the Cauchy criterion for sequences of real numbers, we can easily see that a sequence of complex numbers converges if and only if it is a Cauchy sequence.

Lemma 2.5 (Convergence of Cauchy Series over $\mathbb{C}$)

Interpreting the sum of a series of complex numbers

$$z_1 + z_2 + \ldots + z_n + \ldots \tag{46}$$

as the limit of the sequence its partial sums $\{s_n\}$, where $s_n = z_1 + \ldots z_n$ as $n \to \infty$, we can see that the series converges if and only if for every $\epsilon > 0$ there exists a $N \in \mathbb{N}$ such that

$$|z_m + \ldots + z_n| < \epsilon \tag{47}$$

for any natural numbers $n \geq m > N$.

Definition 2.5 (Absolute Convergence of $\mathbb{C}$)

A series $z_1 + \ldots + z_n + \ldots$ of complex numbers is **absolutely convergent** if the series

$$|z_1| + |z_2| + \ldots + |z_n| + \ldots \tag{48}$$

converges. Clearly, is a series converges absolutely, then it converges due to the inequality

$$|z_m + \ldots + z_n| \leq |z_m| + \ldots + |z_n| \tag{49}$$

Example 2.1 (Absolutely Convergent Complex Series)

The following complex series converge because they converge absolutely. That is,

$$1 + \frac{1}{1!}|z| + \frac{1}{2!}|z^2| + \ldots \text{ converges } \forall\, \mathbb{C} \implies 1 + \frac{1}{1!}z + \frac{1}{2!}z^2 + \ldots \text{ converges } \forall\, \mathbb{C}$$
$$|z| + \frac{1}{3!}|z|^3 + \frac{1}{5!}|z|^5 + \ldots \text{ converges } \forall\, \mathbb{C} \implies z - \frac{1}{3!}z^3 + \frac{1}{5!}z^5 + \ldots \text{ converges } \forall\, \mathbb{C}$$
$$1 + \frac{1}{2!}|z|^2 + \frac{1}{4!}|z|^4 + \ldots \text{ converges } \forall\, \mathbb{C} \implies 1 - \frac{1}{2!}z^2 + \frac{1}{4!}z^4 - \ldots \text{ converges } \forall\, \mathbb{C}$$

Definition 2.6 (Complex Power Series)

Series of the form

$$\sum_{n=0}^{\infty} c_n(z - z_0)^n = c_0 + c_1(z - z_0) + \ldots + c_n(z - z_0) + \ldots \tag{50}$$

are called **complex power series**, or **power series over** $\mathbb{C}$.

But a power series is quite useless unless we know the domain in which is converges (again, note that it is not always guaranteed to converge onto the function f if its power series expansion does converge at all). To develop more sophisticated tests of convergence of a complex power series, we introduce the complex analogue of the root test for real power series.

Theorem 2.6 (Cauchy-Hadamard Theorem)

The complex power series

$$c_0 + c_1(z - z_0) + \ldots + c_n(z - z_0) + \ldots \tag{51}$$

converges inside the disk $|z - z_0| < R$ with center at z_0 and radius given by the formula

$$R = \frac{1}{\overline{\lim}_{n\to\infty} \sqrt[n]{|c_n|}} = \frac{1}{\lim_{n\to\infty} \sup \sqrt[n]{|c_n|}} \tag{52}$$

Where $\overline{\lim}$ denotes the superior limit. Furthermore,

1. the power series diverges at any point exterior to the disk.
2. the power series converges absolutely at any point interior to the disk.
3. the power series is indeterminate at any point on the boundary of the disk.

Note that in the degenerate case when $R = 0$, the series converges only at the point $z = z_0$.

Corollary 2.7 (Abel's First Theorem on Power Series)

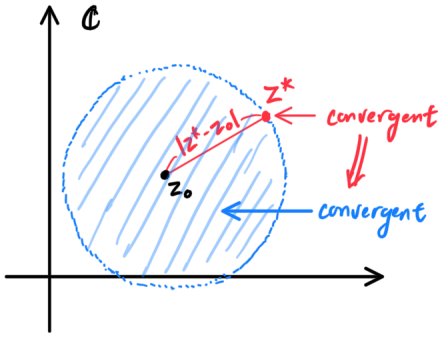
If the power series

$$c_0 + c_1(z - z_0) + \ldots + c_n(z - z_0) + \ldots \tag{53}$$

converges at some value z^*, then it converges absolutely for any value of z satisfying

$$|z - z_0| < |z^* - z_0| \tag{54}$$

The values of z satisfying the inequality above can be intuitively visualized as the following region.



Theorem 2.8 (Product of Absolutely Convergent Series)

Let $a_1 + a_2 + \ldots$ and $b_1 + b_2 + \ldots$ be an absolutely convergent series such that

$$\sum_{i=1}^{\infty} a_i = A \text{ and } \sum_{j=1}^{\infty} b_j = B \tag{55}$$

Then, the Cauchy product of the two series

$$\left(\sum_{i=1}^{\infty} a_i\right) \cdot \left(\sum_{j=1}^{\infty} b_j\right) = \sum_{k=0}^{\infty} c_k = AB, \text{ where } c_k = \sum_{l=0}^{k} a_l b_{k-l} \tag{56}$$

$a_1b_1 + a_2b_2 + \ldots$ is absolutely convergent and

$$\sum_{i=1}^{\infty} a_i b_i = AB \tag{57}$$

Proof. To be done.

Example 2.2 (Convergence of the Cauchy Product of Absolutely Convergent Complex Series)

The two series

$$\sum_{n=0}^{\infty} \frac{1}{n!} a^n \text{ and } \sum_{m=0}^{\infty} \frac{1}{m!} b^m \tag{58}$$

converges absolutely. Therefore, we can see that their Cauchy product can be nicely represented by

grouping together all monomials of the form $a^n b^m$ having the same total degree $m+n=k$.

$$\left(\sum_{n=0}^{\infty} \frac{1}{n!}a^n\right) \cdot \left(\sum_{m=0}^{\infty} \frac{1}{m!}b^m\right) = \sum_{k=0}^{\infty}\left(\sum_{n+m=k} \frac{1}{n!}a^n \frac{1}{m!}b^m\right) \tag{59}$$

But we can simplify

$$\sum_{m+n=k} \frac{1}{n!m!}a^n b^m = \frac{1}{k!}\sum_{n=0}^{k} \frac{k!}{n!(k-n)!}a^n b^{k-n} = \frac{1}{k!}(a+b)^k \tag{60}$$

and therefore we find that

$$\left(\sum_{n=0}^{\infty} \frac{1}{n!}a^n\right) \cdot \left(\sum_{m=0}^{\infty} \frac{1}{m!}b^m\right) = \sum_{k=0}^{\infty} \frac{1}{k!}(a+b)^k \tag{61}$$

2.3 Euler's Formula

Definition 2.7 (Complex Taylor Expansions of Transcendental Functions)

Since we have determined absolute convergence, and therefore convergence, of all these series in all of $\mathbb{C}$, it is natural to extend the definitions of

$$\exp, \cos, \sin : \mathbb{R} \to \mathbb{R} \tag{62}$$

to the complex field

$$\exp, \cos, \sin : \mathbb{C} \to \mathbb{C} \tag{63}$$

by defining them as

$$\begin{aligned} e^z &\equiv 1 + \frac{1}{1!}z + \frac{1}{2!}z^2 + \frac{1}{3!}z^3 + \dots \\ \cos z &\equiv 1 - \frac{1}{2!}z^2 + \frac{1}{4!}z^4 - \frac{1}{6!}z^6 + \dots \\ \sin z &\equiv z - \frac{1}{3!}z^3 + \frac{1}{5!}z^5 - \frac{1}{7!}z^7 + \dots \end{aligned}$$

Notice that even in the complex field, $\cos z$ is an even function and $\sin z$ is an odd function.

$$\begin{aligned} \cos(-z) &= \cos(z) \\ \sin(-z) &= -\sin(z) \end{aligned}$$

In fact, the last example in the previous subsection just proves the following.

Lemma 2.9 (Exponential Map as a Group Homomorphism)

The exponential map $\exp : \mathbb{C} \to \mathbb{C} \setminus \{0\}$ satisfies the following

$$\exp(z_1 + z_2) = \exp(z_1) \cdot \exp(z_2) \tag{64}$$

That is, exp is a group homomorphism from $(\mathbb{C}, +)$ to $(\mathbb{C} \setminus \{0\}, \cdot)$.

Definition 2.8 (Euler's Formula)

By making the substitution $z = yi$ in the series expansion of e^z (where y is an arbitrary complex number), we get

$$\begin{aligned} e^{iy} &= 1 + \frac{1}{1!}(iy) + \frac{1}{2!}(iy)^2 + \frac{1}{3!}(iy)^3 + \frac{1}{4!}(iy)^4 + \dots \\ &= \left(1 - \frac{1}{2}y^2 + \frac{1}{4!}y^4 - \dots\right) + i\left(\frac{1}{1!}y - \frac{1}{3!}y^3 + \frac{1}{5!}y^5 - \dots\right) \end{aligned}$$

which brings us the identity

$$e^{iy} = \cos y + i \sin y \tag{65}$$

Since cos is even and sin is odd, we can add the two identities

$$\begin{aligned} e^{iz} &= \cos z + i \sin z \\ e^{-iz} &= \cos z - i \sin z \end{aligned}$$

to get

$$\begin{aligned} \cos z &= \frac{1}{2}\left(e^{iz} + e^{-iz}\right) \\ \sin z &= \frac{1}{2i}\left(e^{iz} - e^{-iz}\right) \end{aligned}$$

This gives us a very elegant connection between these three transcendental functions.

Definition 2.9 (Hyperbolic Functions)

Likewise, the following series are convergent (since they are absolutely convergent) and therefore we can define the extension of cosh and sinh into the complex field as

$$\begin{aligned} \cosh z &\equiv 1 + \frac{1}{2!}z^2 + \frac{1}{4!}z^4 + \frac{1}{6!}z^6 + \dots \\ \sinh z &\equiv z + \frac{1}{3!}z^3 + \frac{1}{5!}z^5 + \frac{1}{7!}z^7 + \dots \end{aligned}$$

The following identities immediately follow

$$\begin{aligned} \cosh z &= \frac{1}{2}\left(e^{z} + e^{-z}\right) \\ \sinh z &= \frac{1}{2}\left(e^{z} - e^{-z}\right) \end{aligned}$$

Lemma 2.10 (Trigonometric, Hyperbolic Identities over $\mathbb{C}$)

Common identities, which are exactly the same as their real analogues, are listed.

1. $\cos^2 z + \sin^2 z = 1$
2. $\cosh^2 z - \sinh^2 z = 1$
3. $e^{i(z_1+z_2)} = (\cos z_1 \cos z_2 - \sin z_1 \sin z_2) + i(\sin z_1 \cos z_2 + \cos z_1 \sin z_2)$
4. $\cos(z_1 + z_2) = \cos z_1 \cos z_2 - \sin z_1 \sin z_2$
5. $\sin(z_1 + z_2) = \sin z_1 \cos z_2 + \cos z_1 \sin z_2$
6. $\cosh z = \cos iz$
7. $\sinh z = -i \sin iz$

However, to obtain even such geometrically obvious facts as the equality

$$\sin \pi = 0 \text{ or } \cos z + 2\pi = \cos z \tag{66}$$

from the power series definitions of cos and sin is extremely difficult. What the properties actually do is present the remarkable unity of these seemingly different trigonometric and hyperbolic functions, which would have been impossible to detect without going into the domain of complex numbers.

If we just take the following identities

$$\begin{aligned} \cos x &= \cos(x + 2\pi) \\ \sin x &= \sin(x + 2\pi) \\ \cos 0 &= 1 \\ \sin 0 &= 0 \end{aligned}$$

then we get the following identity.

Theorem 2.11 (Euler's Identity)

The following relation is true.

$$e^{i\pi} + 1 = 0 \tag{67}$$

which immediately implies

$$\exp(z + 2\pi i) = \exp z \tag{68}$$

That is, the exponential function is a periodic function on $\mathbb{C}$ with the purely imaginary period $T = 2\pi i$.

Corollary 2.12 (Trigonometric Notation of Complex Number)

With Euler's formula and the periodic relation of $\exp z$, the trigonometric form of a complex number can be presented as

$$z = r(\cos \varphi + i \sin \varphi) = re^{i\varphi} \tag{69}$$

We can rewrite De Moivre's formula as

$$z^n = r^n e^{n\varphi i} \tag{70}$$

2.4 Continuity, Differentiability, Analyticity of Complex Functions

The definitions of continuity and differentiability are the same, just under a different field.

Definition 2.10 (Limit of a Complex Function)

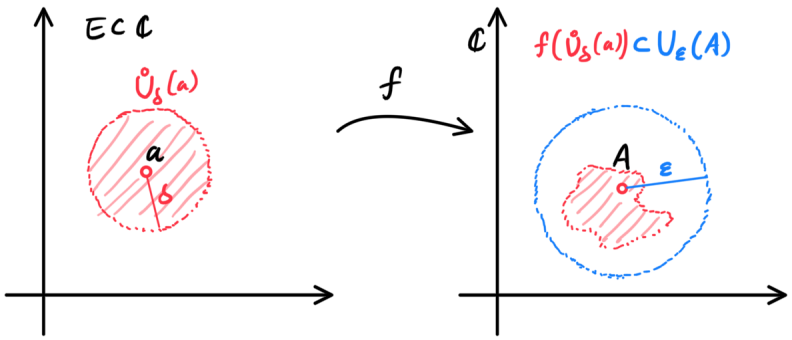
The function $f : E \subset \mathbb{C} \to \mathbb{C}$ tends to $A \in \mathbb{C}$ as $z \to a$, or that

$$\lim_{z \to a} f(z) = A \tag{71}$$

if for every $\epsilon > 0$ there exists a $\delta > 0$ such that

$$0 < |z - a| < \delta \implies |f(z) - A| < \epsilon \tag{72}$$

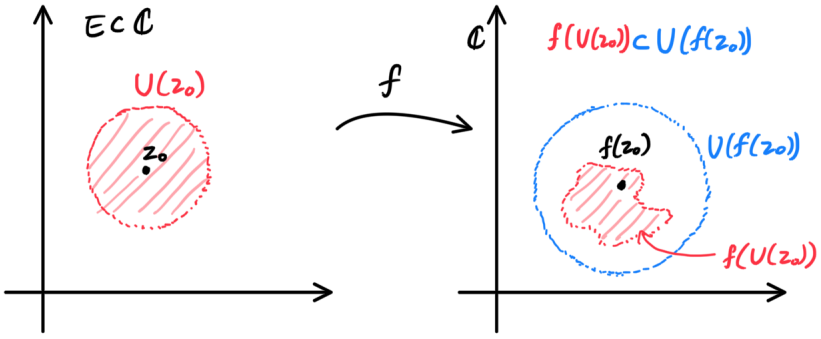
Note that we set $0 < |z - a|$ to ensure that $z \neq a$.
Therefore, in other words, for any arbitrarily small $\epsilon > 0$, we can find a $\delta > 0$ such that the image of the deleted δ-neighborhood of a, denoted $\mathring{U}_\delta(a))$, is completely within the ϵ-neighborhood $U_\epsilon(A)$.



Definition 2.11 (Continuity of a Complex Function)

A function $f : E \subset \mathbb{C} \to \mathbb{C}$ is **continuous** at a point $z_0 \in E$ if for any neighborhood $U(f(z_0))$ there exists a neighborhood $U(z_0)$ such that its image is contained in $U(f(z_0))$. In short,

$$\lim_{z \to z_0} f(z) = f(z_0) \tag{73}$$



Definition 2.12 (Differentiability of a Complex Function)

The **derivative** of a function $f : E \subset \mathbb{C} \to \mathbb{C}$ is defined

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \tag{74}$$

if this limit exists. f **differentiable** at x_0 means that a differential function

$$df(z_0) : T_{z_0}\mathbb{C} \to T_{f(z_0)}\mathbb{C}, \quad h \mapsto df(z_0)(h) \tag{75}$$

exists such that

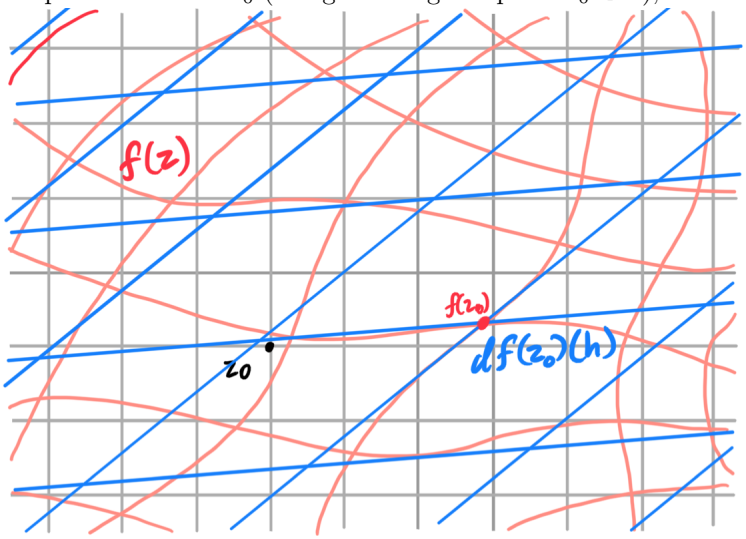
$$f(z) = f(z_0) + df(z_0)(h) + o(h) \tag{76}$$

where $h = z - z_0$ is the increment of the argument. Just like the real case, it turns out that $df(z_0)(h) = f'(z_0)h$, and

$$f(z) - f(z_0) = f'(z_0)(z - z_0) + o(z - z_0) \tag{77}$$

which elegantly weaves together the two concepts of differentiability and the derivative.

Visualizing this, we can see that for whatever function $f : \mathbb{C} \to \mathbb{C}$ there is a linear function that transforms the entire space as such at z_0 (along with a given point $z_0 \in \mathbb{C}$),



The differential $df(z_0)$ at the point z_0 is a linear mapping that "best" approximates f, with an error of $o(h) = o(z - z_0)$.

Lemma 2.13 (Arithmetic Properties of Differentiation over $\mathbb{C}$)

If functions $f, g : E \subset \mathbb{C} \to \mathbb{C}$ are differentiable at a point $z \in E$, then

1. their sum is differentiable at z, and

$$d(f+g)(z) = df(z) + dg(z) \iff (f+g)'(z) = (f' + g')(z) \tag{78}$$

2. their product is differentiable at z, and

$$d(f \cdot g)(z) = g(z)df(z) + f(z)dg(z) \iff (f \cdot g)'(z) = f'(z)g(z) + f(z) \cdot g'(z) \tag{79}$$

3. their quotient is differentiable at z if $g(z) \neq 0$, and

$$d\left(\frac{f}{g}\right)(z) = \frac{g(z)df(z) - f(z)dg(z)}{g^2(z)} \iff \left(\frac{f}{g}\right)'(z) = \frac{f'(z)g(z) - f(z)g'(z)}{g^2(z)} \tag{80}$$

Just like the real case, the operation of taking the derivative is a linear operator.

Lemma 2.14 (Chain Rule for Composite Functions over $\mathbb{C}$)

Let there be functions $f : E_1 \subset \mathbb{C} \to \mathbb{C}$ differentiable at point $z \in E_1$ and $g : E_2 \subset \mathbb{C} \to \mathbb{C}$ differentiable at point $w = f(z) \in E_2$, with respective differentials

$$df(z) : T_z\mathbb{C} \to T_w\mathbb{C}$$
$$dg(w) : T_w\mathbb{C} \to T_{g(w)}\mathbb{C}$$

Then, the composite function $g \circ f : E_1 \to \mathbb{C}$ is differentiable at z, and $d(g \circ f)(z) : T_z\mathbb{C} \to T_{g \circ f(z)}\mathbb{C}$ is

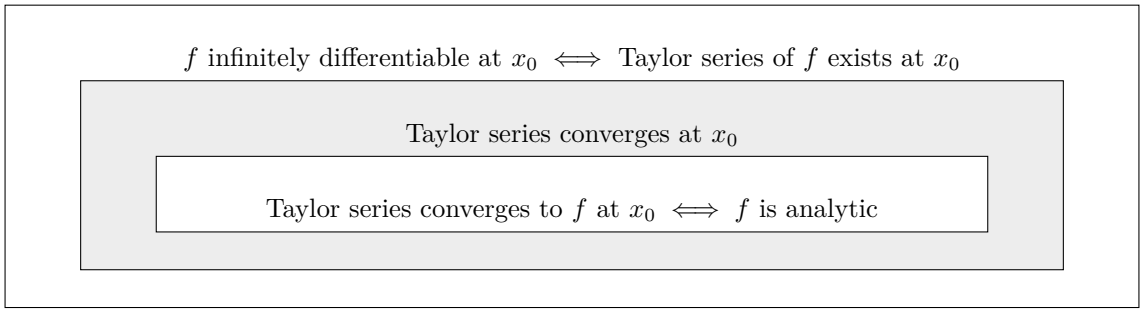
$$d(g \circ f)(z) = dg(w) \circ df(z) \iff (g \circ f)'(z) = g'\big(f(z)\big) \circ f'(z) \tag{81}$$

2.4.1 Power Series Representation of a Function

Definition 2.13 (Holomorphic Function)

If function $f : E \subset \mathbb{C} \to \mathbb{C}$ is (complex) differentiable at a point $z_0 \in E$, then f is said to be **holomorphic at** z_0.

We recall the diagram that summarizes the conditions of differetiability and analyticity of a function f over the field $\mathbb{R}$.



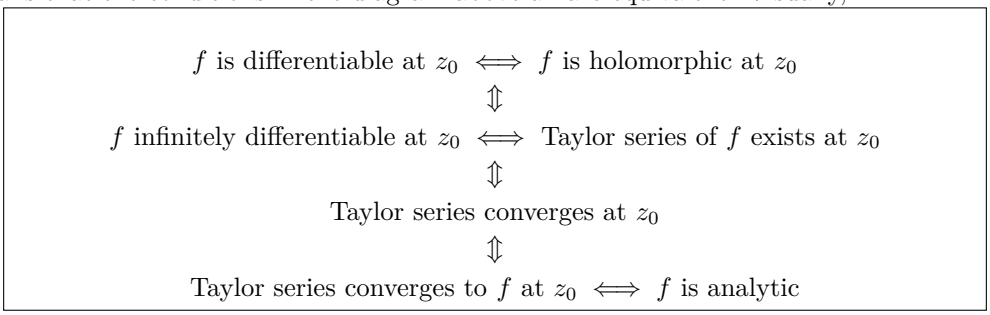
In the theory of functions of a complex variable we actually have a remarkable theorem that does not have an analogue for functions over $\mathbb{R}$.

Theorem 2.15 (Analyticity of Differentiable Functions over $\mathbb{C}$)

If a function $f : E \subset \mathbb{C} \to \mathbb{C}$ is differentiable in a neighborhood of a point $z_0 \in E$, then it is analytic at that point. In other words,

$$f \text{ is holomorphic at } z_0 \implies f \text{ is analytic at } z_0 \tag{82}$$

This means that the conditions in the diagram above all are equivalent! Visually,



This is certainly an amazing fact, since it then follows from the theorem that if a function $f(z)$ has one derivative $f'(z)$ in a neighborhood of a point, it also has derivatives of all orders in that neighborhood.

2.4.2 Algebraic Closedness of the Field $\mathbb{C}$

Definition 2.14 (Algebraically Closed Field)

A field $\mathbb{F}$ is **algebraically closed** if every nonconstant polynomial in $\mathbb{F}[x]$ (the polynomial ring with coefficients in $\mathbb{F}$) has a root in $\mathbb{F}$.

Theorem 2.16 (Fundamental Theorem of Algebra)

$\mathbb{C}$ is algebraically closed. That is, every polynomial

$$P(z) \equiv c_0 + c_1 z + c_2 z^2 + \ldots + c_n z^n \tag{83}$$

of degree $n \geq 1$ with complex coefficients $c_i \in \mathbb{C}$ $(i = 0, 1, \ldots, n)$ has a root in $\mathbb{C}$. This immediately implies that every polynomial $P(z)$ admits a representation (unique up to the order of the factors) in the form

$$P(z) = c_n(z - z_1)(z - z_2)\ldots(z - z_n) \tag{84}$$

where $z_1, \ldots, z_n \in \mathbb{C}$ not necessarily all distinct.

We can also prove the interesting property about zeroes of polynomials in $\mathbb{R}[x]$.

Corollary 2.17 (Complex Conjugate Roots of Real Polynomials)

Given a polynomial with real coefficients

$$P(z) \equiv a_0 + a_1 z + a_2 z^2 + \ldots + a_n z^n \tag{85}$$

P, as we know, does not always have real roots (e.g. $P(x) = x^2 + 1$). However, we state that

$$\text{if } P(z_0) = 0, \text{ then } P(\overline{z}_0) = 0 \tag{86}$$

Therefore, every polynomial P with real coefficients can be expanded as a product of linear and quadratic polynomial with real coefficients.

Proof. We can see from the properties of complex numbers that

$$\begin{aligned}
\overline{(z_1 + z_2)} &= \overline{z_1} + \overline{z_2} \\
\overline{(z_1 \cdot z_2)} &= \overline{(r_1 e^{i\varphi_1} \cdot r_2 e^{i\varphi_2})} \\
&= \overline{r_1 r_2 e^{i(\varphi_1 + \varphi_2)}} = r_1 r_2 e^{-i(\varphi_1 + \varphi_2)} \\
&= r_1 e^{-i\varphi_1} \cdot r_2 e^{-i\varphi_2} = \overline{z}_1 \cdot \overline{z}_2
\end{aligned}$$

Thus, if $P(z_0) = 0$, then

$$0 = \overline{P(z_0)} = \overline{a_0 + \ldots + a_n z_0^n} = \overline{a}_0 + \ldots + \overline{a}_n \overline{z}_0^n = a_0 + \ldots + a_n \overline{z}_0^n = P(\overline{z}_0) \tag{87}$$

and thus $P(\overline{z}_0) = 0$.

2.5 Primitives

Definition 2.15 (Primitive)

A function $F(x)$ is a **primitive** of a function $f(x)$ on an interval if F is differentiable on the interval and satisfies the equation

$$F'(x) = f(x) \tag{88}$$

or equivalently, if their respective differentials satisfy

$$dF(x) = f(x)\, dx \tag{89}$$

Lemma 2.18 (Primitives Differ by a Constant)

If $F_1(x)$ and $F_2(x)$ are two primitives of $f(x)$ on the same interval, then the difference $(F_1 - F_2)(x)$ is constant on that interval.

Example 2.3 ($\arctan x$ and $\operatorname{arccot}(1/x)$ as Primitives)

Both

$$F_1(x) \equiv \arctan x \text{ and } F_2(x) \equiv \operatorname{arccot} \frac{1}{x} \tag{90}$$

are primitives of $f(x) = \frac{1}{1+x^2}$. Indeed, we can see by direct calculation that in the domain $\mathbb{R} \setminus 0$,

$$F_1(x) - F_2(x) = \arctan x - \operatorname{arccot} \frac{1}{x} = \begin{cases} 0, & x > 0 \\ -\pi, & x < 0 \end{cases} \tag{91}$$

which is supported by the lemma.

Notice how given a function $f(x)$, the operation of finding its differential, denoted with d, gives us a new function of h, called the differential

$$df(x)(h) \tag{92}$$

Similarly, the operation of finding a primitive of function $f(x)$, denoted with the symbol $\int$, gives us a new function.

Definition 2.16 (Indefinite Integration)

The operation of finding a primitive of a certain function $f(x)$ is called **indefinite integration**, and the mathematical notation

$$\int f(x)\,dx \tag{93}$$

is called the **indefinite integral of** $f(x)$ on a given interval (f called the **integrand** and $f(x)\,dx$ called the **differential form**).

1. It immediately follows from the lemma that if $F(x)$ is any particular primitive of $f(x)$ on the interval, then on that interval
$$\int f(x)\,dx = F(x) + C \tag{94}$$
2. If $F'(x) = f(x)$ (that is, F is a primitive of f on some interval), then we have
$$d \int f(x)\,dx = dF(x) = F'(x)\,dx \tag{95}$$
3. It also follows that
$$\int dF(x) = \int F'(x)\,dx = F(x) + C \tag{96}$$

Theorem 2.19 (Basic Methods of Indefinite Integration)

The definition of the indefinite integral has three basic properties that can be used to solve indefinite integrals.

1. Linearity of the indefinite integral.

$$\int \big(\alpha u(x) + \beta v(x)\big)\, dx = \alpha \int u(x)\, dx + \beta \int v(x)\, dx + C \tag{97}$$

2. Integration by parts.

$$\int (uv)'\, dx = \int u'(x)v(x)\, dx + \int u(x)v'(x)\, dx + C \tag{98}$$

3. Change of Variable, or U-substitution. Given that $F'(x) = f(x)$ on an interval I_x and $\varphi : I_t \to I_x$ is a C^1 mapping of interval I_t into I_x, then

$$\int (f \circ \varphi)(t)\varphi'(t)\, dt = (F \circ \varphi)(t) + C \tag{99}$$

3 Jan 8

We start off with some useful geometry, which doesn't seem relevant now but will be helpful later on.

Definition 3.1 (Complex Projective Line)

Define equivalence relation on $\mathbb{C}^2 \setminus \{0\}$ by specifiying $\zeta \sim \xi$ iff $\exists 0 = c \in \mathbb{C}$ s.t. $\xi = c\zeta$. Here, $0 \neq \zeta, \xi \in \mathbb{C}^2$.
That is, $\xi \sim \zeta$ means that they span the same 1-dimensional linear subspace. Then, we define the **complex projective line** as

$$\mathbb{CP}^1 = (\mathbb{C}^2 \setminus \{0\})/\sim \tag{100}$$

The $\mathbb{C}$ stands for complex, but we will omit it since this is about complex analysis. The set of all 1-dimensional linear subspaces is called the **Grassmanian**, denoted

$$\mathrm{Gr}(1, \mathbb{C}^2) \tag{101}$$

These spaces are all the same.

For notation, given $\zeta = (\zeta_0, \zeta_1) \neq 0 \in \mathbb{C}^2$, we denote $[\zeta] = [\zeta_0 : \zeta_1] \in \mathbb{P}^1$ be the line spanned by ζ.

A philosophy of complex geometry is that to study a space, you should study the functions on that space. So we will study *holomorphic functions* on the space. So what are the functions are $\mathbb{P}^1$? We care about holomorphic functions, and the only holomorphic functions on $\mathbb{P}^1$ tend to be complex.

Let's let $\mathbb{C}[\zeta_0, \zeta_1]$ the ring of complex polynomials in two variables, and let $\mathbb{C}[\zeta_0, \zeta_1]_d \subset \mathbb{C}[\zeta_0, \zeta_1]$ be the vector space of homogeneous polynomials of degree d.

Suppose $P \in \mathbb{C}[\zeta_0, \zeta_1]$. Then P is homogeneous of degree d iff $P(\lambda\zeta_0, \lambda\zeta_1) = \lambda^d P(\zeta_0, \zeta_1)$ for all $\zeta \in (\zeta_0, \zeta_1) \in \mathbb{C}^2$ and for all $\lambda \in \mathbb{C}$.

Fix a line $[\zeta] \in \mathbb{P}^1$ in $\mathbb{C}^2$. This line is paramterized by $\lambda \in \mathbb{C} \to \lambda\zeta \in \mathbb{C}^2$. If we restrict P to this line, it takes the values $P(\lambda\zeta) = \lambda^d P(\zeta)$. Therefore, P does *not* define a function on $\mathbb{P}^1$. Intuitively, we can think of this as "if it vanishes at 1 (nonzero) point on the line, it vanishes at all points on the line." But $\{P = 0\} \subset \mathbb{P}^1$ is well-defined, because $P(\zeta) = 0 \implies P(c\zeta) = 0$ for all $c \in \mathbb{C} \implies P(\xi) = 0$ for all $\xi \sim \zeta$.

Theorem 3.1 (Homogeneous Polynomials Induce Maps of $\mathbb{P}^1$)

Suppose $P, Q \in \mathbb{C}[\zeta_0, \zeta_1]_d$. We claim the map $F : \mathbb{P}^1 \to \mathbb{P}^1$ defined by

$$[\zeta] \mapsto [P(\zeta), Q(\zeta)] \tag{102}$$

is well-defined[a] as long as $\{P = 0\} \cap \{Q = 0\} = \{0\} \subset \mathbb{C}^2$.

[a] Maps of this form are called *rational functions*, which has some connection to what you already might think of as rational functions.

Proof. Suppose $\zeta \neq 0 \in \mathbb{C}^2$, but $P(\zeta), Q(\zeta) = 0$, i.e. is in the 0-locus of both polynomials. Then $[P(\zeta) : Q(\zeta)] = [0 : 0]$ is not defined. So $\{P = 0\} \cap \{Q = 0\} = \{0\}$ is necessary.
To see that F is well-defined, we must show that F takes the same value on $[\zeta]$ and $[c\zeta]$ for all $0 \neq c \in \mathbb{C}$.

$$\begin{aligned} F([c\zeta]) &= [P(c\zeta) : Q(c\zeta)] && (103)\\ &= [c^d P(\zeta) : c^d Q(\zeta)] && (104)\\ &= [P(\zeta) : Q(\zeta)] = F([\zeta]) && (105) \end{aligned}$$

So the definition of F is independent of the generator of the line, so this is nice.

Example 3.1 ($\mathrm{GL}_2\mathbb{C}$ Acts on $\mathbb{P}^1$)

If $A \in \mathrm{GL}_2\mathbb{C}$, then

$$[\zeta] \mapsto [A\zeta] \tag{106}$$

is a well-defined map $[A] : \mathbb{P}^1 \to \mathbb{P}^1$. Maps of this form are called Mobius transformations.[a] Just do the same thing.

[a] very important since they are very flexible and can be used to normalize problems to make solutions much easier.

We know that $\mathbb{P}^1$ is the 1-point compactification of $\mathbb{C}$. We can write $\mathbb{P}^1 = U_0 \cup U_1$, where

$$U_0 = \{[\zeta_0 : \zeta_1] \in \mathbb{P}^1 \colon \zeta_0 \neq 0\} \tag{107}$$

$$U_1 = \{[\zeta_0 : \zeta_1] \in \mathbb{P}^1 \colon \zeta_1 \neq 0\} \tag{108}$$

This is handy, since it gives me a sneaky way to put the complex plane into the projective line. Define map $\mathbb{C} \to U_1$,

$$z \mapsto [z : 1] \in U_1 \subset \mathbb{P}^1 \tag{109}$$

This maps is bijective, with inverse

$$[\zeta_0 : \zeta_1] = \left[\frac{\zeta_0}{\zeta_1} : 1\right] \mapsto \frac{\zeta_0}{\zeta_1} = z \tag{110}$$

What we missed is

$$\mathbb{P}^1 \setminus U_1 = \{[\zeta_0 : 0] \colon \zeta_0 \neq 0\} \tag{111}$$

$$= \{[1 : 0]\} \tag{112}$$

which is just one point. We call $[1 : 0] = \infty$, i.e. the point at infinity. So we have

$$\mathbb{P}^1 = U_1 \sqcup \{[1 : 0]\} = \mathbb{C} \cup \{\infty\} \tag{113}$$

Conway calls this space $\mathbb{C}_\infty$, but we also call is $\mathbb{R} \cup \{\infty\} = \mathbb{RP}^1$. But to talk about a compactification, we must extend the topology of $\mathbb{C}$ to $\mathbb{P}^1$ by declaring

$$B(\infty, \frac{1}{r}) = \{z \in \mathbb{C} \mid |z| > r\} \cup \{\infty\} \tag{114}$$

to be in the basis of $+\infty = [1 : 0] \in \mathbb{P}^1$.

Exercise 3.1 (Continuity of Rational Maps on $\mathbb{P}^1$)

Show that rational functions $F = [P : Q] : \mathbb{P}^1 \to \mathbb{P}^1$ are continuous.

Finally, $\mathbb{P}^1$ is also known as the Riemann sphere because $\mathbb{P}^1$ is homeomorphic to S^2. We can identify $\mathbb{R}^3$ with $\mathbb{C} \times \mathbb{R}$, embed S^2 in $\mathbb{C} \times \mathbb{R}$ s.t.

$$S^2 = \{(z, t) \in \mathbb{C} \times \mathbb{R} \colon |z|^2 + |t|^2 = 1\} \tag{115}$$

and the map is

$$\phi(z) = \left(\frac{2z}{|z|^2 + 1}, \frac{|z|^2 - 1}{|z|^2 + 1}\right) \in S^2 \subset \mathbb{C} \times \mathbb{R} \tag{116}$$

3.1 Rational Functions

Suppose $P \in \mathbb{C}[\zeta_0 : \zeta_1]_d$. Then $p(z) = P(z, 1) \in \mathbb{C}[z]$ is a polynomial of degree $\leq d$. Now conversely, suppose that $p(z) \in \mathbb{C}[z]$ is a polynomial of degree d. We get a homogeneous polynomial of degree d in $\mathbb{C}[\zeta_0, \zeta_1]$ by a process called "homoegenization."

$$P(\zeta) = P(\zeta_0, \zeta_1) = \zeta_1^d p(\zeta_0/\zeta_1) \tag{117}$$

If $p(z) = \sum_{j=0}^d a_j z^j$. Then

$$P(\zeta) = \zeta_1^d \sum_{j=0}^d a_j \Big(\frac{\zeta_0}{\zeta_1}\Big)^j = \sum_{j=0}^d a_j \zeta_0^j \zeta_1^{d-j} \tag{118}$$

So we have a way of toggling back and forth between two variable homogenous polynomials and one variable nonhomogeneous polynomials.

Given $P, Q \in \mathbb{C}[\zeta_0, \zeta_1]_d$, we have the map $F = [P : Q] : \mathbb{P}^1 \to \mathbb{P}^1$. Then, we have

$$\begin{array}{ccc} \mathbb{P}^1 & \xrightarrow{F=[P:Q]} & \mathbb{P}^1 \\ {\scriptstyle i}\uparrow & & \\ \mathbb{C} & & \end{array}$$

Figure 1

Notice that $F \circ i : z \in \mathbb{C} \mapsto [z : 1] \mapsto [p(z) : q(z)] = \left[\frac{p(z)}{q(z)} : 1\right]$. When convenient, we will conflate F with $\frac{p(z)}{q(z)}$.

3.2 Mobius Transformations

Let $A \in \mathrm{GL}_2\mathbb{C}$ and let $[A] : \mathbb{P}^1 \to \mathbb{P}^1$ be the Mobius transformation $[\zeta] \to [A\zeta]$. It is not hard to see that

$$[A][B] = [AB] \tag{119}$$

$$[A]^{-1} = [A^{-1}] \tag{120}$$

Therefore, the set of all mobius transformations,

$$\mathrm{PGL}_2\mathbb{C} = \{[A] : \mathbb{P}^1 \to \mathbb{P}^1 \text{ s.t. } A \in \mathrm{GL}_2\mathbb{C}\} \tag{121}$$

is a group. And the map $\mathrm{GL}_2\mathbb{C} \to \mathrm{PGL}_2\mathbb{C}$ is a group homomorphism. The kernel is the scalar multiple of identity maps. Suppose $A \in \ker$, so $[A] : \mathbb{P}^1 \to \mathbb{P}^1$ is the identity, so this means that it preserves every 1-dimensional subspaces of $\mathbb{C}^2$. Therefore, the whole space must be an eigenspace. Therefore,

$$\mathrm{PGL}_2\mathbb{C} = \mathrm{GL}_2\mathbb{C}(C \times I) = \mathrm{SL}_2\mathbb{C}/\{\pm I\} \tag{122}$$

Let's write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2\mathbb{C} \tag{123}$$

Then, the composition $[A] \circ \iota : \mathbb{C} \to \mathbb{P}^1$ sends

$$z \xrightarrow{\iota} [z : 1] \xrightarrow{[A]} [az + b : cz + d] = \left[\frac{az+b}{cz+d} : 1\right] \tag{124}$$

When convenient, we will conflate $[A]$ with $S_A(z) = \frac{az+b}{cz+d}$.

Exercise 3.2 (Mobius Transformation through Three Points)

Suppose ξ_0, ξ_1, ξ_2 are pairwise disjoint points in $\mathbb{P}^1 = \mathbb{C}_\infty$. Then, the Mobius transformation

$$S(z) = \frac{z - \xi_0}{z - \xi_\infty} \cdot \frac{\xi_1 - \xi_\infty}{\xi_1 - \xi_0} \tag{125}$$

maps $\xi_0 \to 0$, $\xi_1 \to 1$, $\xi_\infty \to \infty$. Therefore, given that I choose three such numbers and you choose three such numbers, there is a Mobius transformation that can send my three numbers to your three numbers simply by mapping it through $0, 1, \infty$. Show that

1. This Mobius transformation satisfying $\xi_0 \to 0$, $\xi_1 \to 1$, $\xi_\infty \to \infty$ is unique. There are many ways to approach this, with the alegebraic argument a bit tedious, but the geometric is another way.
2. Suppose a Mobius transformation T fixes 3 distinct points in $\mathbb{C}_\infty$. Show that T is the identity. This is a straightforward consequence of (a).

4 Jan 13

Recall that

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, [A] : \mathbb{P}^1 \to \mathbb{P}^1 \tag{126}$$

defines a Mobius transformation by $[\xi] \mapsto [A\xi]$. Also, given $\iota : \mathbb{C} \to \mathbb{P}^1$ defined $z \mapsto [z : 1]$ given by $[A] \circ \iota(z) = [az + b : cz + d] = \left[\frac{az+b}{cz+d} : 1\right]$.

We will study actions of Mobius transformations on lines and circles. We will declare lines to be circles of infinite radius. There is a unique circle passing through 3 distinct points $\xi_0, \xi_1, \xi_2 \in \mathbb{P}^1 = \mathbb{C}_\infty$. If one of the points is ∞ or all three points are colinear, then we have a line.

Exercise 4.1 (Mobius Transformations Map Circles to Circles)

Every Mobius transformation maps circles to circles. This is pretty surprising statement, and is surprisingly easy to prove. First show that every Mobius transformation $S(z) = \frac{az+b}{cz+d}$ can be written as a composition of

1. homotheties: $z \mapsto rz$ for $0 < r \in \mathbb{R}$.
2. rotations: $z \mapsto e^{i\theta z}$ for $\theta \in \mathbb{R}$.
3. translations: $z \mapsto z + p$ for $p \in \mathbb{C}$.
4. inversion: $z \mapsto \frac{1}{z}$.

The first three clearly sends circles to circles, but for inversions, you can show it with algebraic calculations.

Corollary 4.1 (Mobius Transformation through Three Points)

The Mobius transformation

$$S(z) = \frac{z - \xi_0}{z - \xi_\infty} \frac{\xi_1 - \xi_\infty}{\xi_1 - \xi_0} \tag{127}$$

maps the circle through $\xi_0, \xi_1, \xi_\infty \in \mathbb{C}_\infty$ to the circle $\mathbb{R}_\infty \subset \mathbb{C}_\infty$ (real line).

Corollary 4.2 (Transitivity on Triples of Points)

The group $\mathrm{PGL}_2(\mathbb{C})$ of Mobius transformations acts transitively on the set of circles.

Proof.

Corollary 4.3 (Cross-Ratio Criterion for Concyclicity)

A point z lies on the circle through $\xi_0, \xi_1, \xi_2 \in \mathbb{C}_\infty$ iff

$$\underbrace{(z, \xi_1, \xi_0, \xi_\infty)}_{\text{cross ratio}} = \frac{z-\xi_0}{z-\xi_\infty}\frac{\xi_1-\xi_\infty}{\xi_1-\xi_0} \in \mathbb{R}_\infty \tag{128}$$

We will use Mobius transformations to analyze functions.

Theorem 4.4 (Mobius Transformations Preserving the Unit Disc)

The subgroup of Mobius transformations preserving the disc $B = \{z \in \mathbb{C} \colon |z| < 1\}$ acts transitively on B, i.e. given $w, w' \in B$, there exists a Mobius transformation $S(z)$ s.t. $S(B) = B$, $S(w) = w'$.

Proof. Very messy to do directly, but becomes easy when we think of it as linear maps. Define a Hermitian form ϕ in $\mathbb{C}^2$ by

$$\phi(\zeta, \xi) = -\zeta_0\bar{\xi_0} + \zeta_1\bar{\xi_1}, \quad \zeta = \begin{bmatrix}\zeta_0 \\ \zeta_1\end{bmatrix}, \xi = \begin{bmatrix}\xi_0 \\ \xi_1\end{bmatrix} \in \mathbb{C}^2 \tag{129}$$

We can set

$$H = \begin{bmatrix}-1 & 0 \\ 0 & 1\end{bmatrix} \tag{130}$$

Then, $h(\zeta, \xi) = \zeta^T H \bar{\xi}$. The automorphism group of ϕ is

$$\begin{aligned}\mathrm{GL}_2(\mathbb{C}, \phi) &= \{A \in \mathrm{GL}_2(\mathbb{C}) \colon \phi(\zeta, \xi) = \phi(A\zeta, A\xi) \forall \zeta, \xi \in \mathbb{C}^2\} && (131)\\ &= \{A \in \mathrm{GL}_2(\mathbb{C}) \colon A^T H \bar{A} = H\} && (132)\\ &\simeq U(1,1) && (133)\end{aligned}$$

To write it out,

$$A = \begin{bmatrix}a & b \\ c & d\end{bmatrix} = \begin{bmatrix}\zeta & \xi\end{bmatrix} \tag{134}$$

Then we have the characterization

$$A \in \mathrm{GL}_2(\mathbb{C}, \phi) \iff \begin{cases}\phi(\zeta, \zeta) &= -1 \\ \phi(\xi, \xi) &= 1 \\ \phi(\zeta, \xi) &= 0\end{cases} \tag{135}$$

For sake of argument, let's write $\zeta = (z, 1)^T$. Then $\phi(\zeta, \zeta) = -|z|^2 + 1$. Then we can identify $\mathbb{P}^1 = B \sqcup S' \sqcup B^*$, where

$$\begin{aligned}B &= \{[\zeta] \in \mathbb{P}^1 \colon \phi(\zeta, \zeta) > 0\} = \{[z:1] \in \mathbb{P}^1 \colon |z| < 1\} && (136)\\ S^1 = \partial B &= \{[\zeta] \in \mathbb{P}^1 \colon \phi(\zeta, \zeta) = 0\} = \{[z:1] \in \mathbb{P}^1 \colon |z| = 1\} && (137)\\ B^* &= \{[\zeta] \in \mathbb{P}^1 \colon \phi(\zeta, \zeta) < 0\} = \{[z:1] \in \mathbb{P}^1 \colon |z| > 1\} && (138)\\ &= \{[1:z] \in \mathbb{P}^1 \colon |z| < 1\} && (139)\end{aligned}$$

So every matrix gives me a Mobius transformation. By construction, if the transformation preserves ϕ, it preserves each of these sets. That is, if $A \in \mathrm{GL}_2(\mathbb{C}, \phi)$, then the Mobius transformation $[A] : \mathbb{P}^2 \to \mathbb{P}^2$ preserves B, S^1, B^*.

Fix $w, w' \in B$. It remains to find $A \in \mathrm{GL}_2(\mathbb{C}, \phi)$ s.t. $S_A(w) = w'$. Fix $\xi = [w : 1] \in B$ s.t. $\phi(\xi, \xi) = 1$. $\zeta \in [1 : w] \in B^*$ s.t. $\phi(\zeta, \zeta) = -1$. Observe that $\phi(\zeta, \xi) = 0 \implies A = [\zeta, \xi] \in \mathrm{GL}_2(A, \phi)$. Similarly, fix $\xi' \in [w' : 1] \in B, \zeta' \in [1 : \bar{w}'] \in B^*$. Then

$$\begin{cases} \phi(\xi', \xi') & = 1 \\ \phi(\zeta', \zeta') & = -1 \implies A' = [\zeta', \xi'] \in \mathrm{GL}_2(\mathbb{C}, \phi) \\ \phi(\zeta, \xi) & = 0 \end{cases} \tag{140}$$

Therefore, $\theta = [0 : 1] \in B$.

$$A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \xi, A' \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \xi' \implies B = A' \circ A^{-1} \text{ maps } \xi \mapsto \xi' \tag{141}$$

which implies that $S_B(w) = w'$.

Definition 4.1 (Differentiable)

Let $U \subset \mathbb{C}$ be open and $p \in U$. Then $f : U \subset \mathbb{C} \to \mathbb{C}$ is **(complex) differentiable at** p if the limit

$$f'(p) \coloneqq \lim_{z \to p} \frac{f(z) - f(p)}{z - p} = \lim_{z \to 0} \frac{f(p + z) - f(p)}{z} \tag{142}$$

exists.

Note that this is much stronger than $\mathbb{R}$-differentiability, since it shouldn't depend on the direction it approaches.

Example 4.1 (Derivatives of Elementary Functions)

1. *Constant.* If $f(z) = c$ is constant, then $f(z) - f(p) = 0 \implies f'(p) = 0$ for all $p \in \mathbb{C}$.
2. *Identity.* If $f(z) = z$, then
$$\frac{f(z) - f(p)}{z - p} = \frac{z - p}{z - p} = 1 \tag{143}$$
so $f'(p) = 1$ for all $p \in \mathbb{C}$.

Theorem 4.5 (Algebraic Rules for $\mathbb{C}$-Differentiation)

Suppose f, g are $\mathbb{C}$-differentiable at $p \in \mathbb{C}$. Then

1. $f + g$ is differentiable, with
$$(f + g)'(p) = f'(p) + g'(p) \tag{144}$$
2. fg is differentiable, with
$$(fg)'(p) = f'(p)g(p) + f(p)g'(p) \tag{145}$$
3. If $g(p) \neq 0$, then f/g is differentiable at p with
$$(f/g)'(p) = \frac{f'(p)g(p) - f(p)g'(p)}{g(p)^2} \tag{146}$$

Proof. Same as in real analysis.

Corollary 4.6 (Polynomials and Rational Functions Are Differentiable)

Polynomials $f(z) \in \mathbb{C}[z]$ are $\mathbb{C}$-differentiable at every point $p \in \mathbb{C}$. Likewise, so are rational functions.

Example 4.2 ($|z|^2$ Is Differentiable Only at the Origin)

$f(z) = |z|^2 = z\bar{z}$ is complex differentiable only at $p = 0 \in \mathbb{C}$.

Note that if f is $\mathbb{C}$ differentiable at p, then there is a "Taylor formula" saying that

$$f(p+z) = f(p) + f'(p)z + \epsilon(z) \tag{147}$$

where $\frac{\epsilon(z)}{z} \to 0$ as $z \to 0$. The Lagrange's form may not be as well-defined nor as useful, but will have to look into this.

Let's compare complex differentiability to real differentiability. Write

$$f(x,y) = u(x,y) + iv(x,y), \quad z = x + iy \in \mathbb{C} \tag{148}$$

Think of f as a map $f : U \subset \mathbb{R}^2 \to \mathbb{R}^2$. Then f is $\mathbb{R}$-differentiable at p if the $\mathbb{R}$ derivative at p exists. If it exists the $\mathbb{R}$-derivative is the unique linear map $Df(p) : \mathbb{R}^2 \to \mathbb{R}^2$ s.t.

$$\lim_{(x,y)\to(0,0)} \frac{\|f(p+(x,y))}{} .. \tag{149}$$

Then as a consequence of the definition, we have Taylor's formula,

$$f(p+(x,y)) = f(p) + Df(p)(x,y) + \epsilon(x,y) \text{ with } \frac{\|\epsilon(x,y)\|}{\|(x,y)\|} \to 0 \text{ as } (x,y) \to 0 \tag{150}$$

So using Taylor's formula, $Df(p)$ has matrix representation (w.r.t. the standard basis).

$$\begin{bmatrix} u_x(p) & v_x(p) \\ u_y(p) & v_y(p) \end{bmatrix} \tag{151}$$

where the subscripts represent partial derivatives. Notice that $x = \frac{1}{2}(z+\bar{z}), y = \frac{i}{2}(\bar{z}-z)$. This motivates

$$\frac{\partial}{\partial z} = \frac{\partial x}{\partial z}\frac{\partial}{\partial x} + \frac{\partial y}{\partial z}\frac{\partial}{\partial y} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right) \tag{152}$$

$$\frac{\partial}{\partial \bar{z}} = \frac{\partial x}{\partial \bar{z}}\frac{\partial}{\partial x} + \frac{\partial y}{\partial \bar{z}}\frac{\partial}{\partial y} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right) \tag{153}$$

where $z, \bar{z} : \mathbb{C} \to \mathbb{C}$. This is motivates and not implies since this isn't a true (strict) change of coordinates.

5 Jan 15

We'll be more concerned with geometric and topological properties rather than ϵ-δ arguments in this class. We have

$$Df(p)\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} xu_x(p) + yu_y(p) \\ xv_x(p) + yv_y(p) \end{pmatrix} \in \mathbb{R}^2 \tag{154}$$

Now think of this as in $\mathbb{C}$, and we get

$$= \big(xu_x(p) + yu_y(p)\big) + i\big(xv_x(p) + yv_y(p)\big) \tag{155}$$

Now we want to write it as in $z, \bar{z}$.

$$= xf_x(p) + yf_y(p) \tag{156}$$

$$= \frac{1}{2}(z+\bar{z})\big(f_z(p) + f_{\bar{z}}(p)\big) + \frac{1}{2}(z-\bar{z})i\big(f_z(p) - f_{\bar{z}}(p)\big) \tag{157}$$

$$= zf_z(p) + \bar{z}f_{\bar{z}}(p) \tag{158}$$

So, using Taylor's formula,

$$f(z+p) = f(p) + zf_z(p) + \bar{z}f_{\bar{z}}(p) + \epsilon(z), \qquad \frac{\epsilon(z)}{z} \to 0 \text{ as } z \to 0 \tag{159}$$

We'll consider 2 cases and show that they're equivalent. Suppose $f_z(p) = 0$. Then,

$$\frac{f(p+z) - f(p)}{z} = f_z(p) + \frac{\epsilon(z)}{z} \implies f'(p) = f_z(p) \text{ exists} \tag{160}$$

so, the complex derivative exists, so the upshot is: $\mathbb{R}$-differentiable and $f_{\bar{z}} = 0$ implies $\mathbb{C}$-differentiable. Conversely, suppose $f'(p)$ exists. Then, writing out

$$f(p+z) = f(p) + f'(p)z + \epsilon(z) \tag{161}$$

defines $\epsilon(z)$, and by complex differentibility, $\frac{\epsilon(z)}{z} \to 0$ as $z \to 0$.

Regard complex multiplication $z \to f'(p)z$ as a $\mathbb{R}$-linear map $\mathbb{R}^2 \to \mathbb{R}^2$. Then, the $\mathbb{R}$-derivative exists, and $f'(p) = f'(p)(x+iy)$ is identified with $Df(p)(x,y)$. Write $f'(p) = \alpha + i\beta$.

$$f'(p)z = (\alpha + i\beta)(x+iy) = (\alpha x - \beta y) + i(\beta x + \alpha y) = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = Df(p) \begin{pmatrix} x \\ y \end{pmatrix} \tag{162}$$

This gives us the Cauchy-Riemann equations

$$\begin{cases} u_x(p) &= \alpha = v_y(p) \\ -u_y(p) &= \beta = v_x(p) \end{cases} \implies \begin{cases} u_x(p) &= v_y(p) \\ u_y(p) &= -v_x(p) \end{cases} \tag{163}$$

Example 5.1 (Cauchy-Riemann via the Wirtinger Derivative)

As an exercise, show that the Cauchy Riemann equations hold iff $f_{\bar{z}}(p) = 0$. Because of this, this is called the complex Cauchy-Riemann equations.

The upshot is: If $f'(p)$ exists, then f is $\mathbb{R}$-differentiable at p and the Cauchy-Riemann equations hold. So we proved in iff statement. So

$$\mathbb{C} - \text{differentiable} \iff \mathbb{R} - \text{differentiable} + \text{CR equations} \tag{164}$$

Here is a consequence geometrically. Suppose $f'(p) = \alpha + i\beta \neq 0$. This implies

$$Df(p) = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} \implies \det Df(p) = \alpha^2 + \beta^2 > 0 \tag{165}$$

$$\implies Df(p) \text{ is orientation preserving} \tag{166}$$

$$\implies \text{complex manifold is always oriented} \tag{167}$$

Odd dimension $\mathbb{RP}$ are orientable, but even-dimensional ones are not orientable. The second consequence is that the columns are orthogonal.

$$Df(p) = \underbrace{\begin{pmatrix} \sqrt{\alpha^2+\beta^2} & 0 \\ 0 & \sqrt{\alpha^2+\beta^2} \end{pmatrix}}_{\text{dilation}} \underbrace{\begin{pmatrix} \frac{\alpha}{\sqrt{\alpha^2+\beta^2}} & \frac{-\beta}{\sqrt{\alpha^2+\beta^2}} \\ \frac{\beta}{\sqrt{\alpha^2+\beta^2}} & \frac{\alpha}{\sqrt{\alpha^2+\beta^2}} \end{pmatrix}}_{\text{orthogonal}} \tag{168}$$

So, $Df(p)$ is a conformal map, i.e. is *angle preserving*.

Definition 5.1 (Holomorphic Function)

Let $U \subset \mathbb{C}$ be open. A function $f : U \to \mathbb{C}$ is **holomorphic** if $f'(p)$ for all $p \in U$.

Example 5.2 (Holomorphy of Polynomials and Rational Functions)

All polynomials are holomorphic. Rational functions are holomorphic on any U on which the denominator does not vanish.

Definition 5.2 (Rings of Holomorphic and Entire Functions)

Let us denote
1. $O(U)$ as the ring of holomorphic functions over U.
2. $O(\mathbb{C})$ as the ring of **entire functions**.

Definition 5.3 (Exponential Map)

Let us define $\exp : \mathbb{C} \to \mathbb{C}$, $z \to e^z = e^{x+iy} = e^x(\cos y + i \sin y)$.

Then, we have

$$\frac{\partial}{\partial \bar{z}} e^z = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)\left(e^x(\cos y + i \sin y)\right) = \frac{1}{2}(e^z - e^z) = 0 \tag{169}$$

which implies that $e^z \in O(\mathbb{C})$. So both exponential and polynomials are entire. It turns out that being complex differentiable implies being analytic.

Theorem 5.1 (Cauchy 1821, Hadamard 1888)

The power series $\sum_{n=0}^{\infty} a_n(z-p)^n$, $a_n \in \mathbb{C}$ converges absolutely and uniformly on $D(p,r) \coloneqq \{z \in \mathbb{C} : |z-p| < r\}$ for all $0 \leq r \leq R$, where

$$R = \frac{1}{\limsup_{n\to+\infty} \sqrt[n]{|a_n|}} \tag{170}$$

and it diverges for every $|z-p| > R$. At equality, it is more subtle—at some points it may diverge or cnverge.

Proof. We prove the two cases.
1. Assume $R > 0$. Fix $0 < r < s < R$. by definition of R as lim sup, there exists $N \in \mathbb{N}$ s.t. $n \geq N \implies |a_n|s^n < 1$. Also,

$$|z-p| < r \implies \sum_{n=N}^{\infty} |a_n||z-p|^n \leq \sum_{n=N}^{\infty} |a_n| r^n = \sum_{n=N}^{\infty} \underbrace{|a_n|s^n}_{<1} \underbrace{\left(\frac{r}{s}\right)^n}_{<1} \tag{171}$$

   By geometric series test, this converges, giving us absolute convergence. Since this doesn't depend on z, this gives us uniform convergence.
2. Assume $R < +\infty$. Fix $R < s < r$. By definition of R, there exists an increasing sequence $S \subset \mathbb{Z}_{>0}$

s.t. $|a_n|s^n > 1$ for all $n \in S$.

$$|z-p| = r \implies |a_n||z-p|^n = \underbrace{|a_n s^n}_{>1, n\in S} \underbrace{\left(\frac{r}{s}\right)^n}_{>1} \implies \text{series diverges} \tag{172}$$

This is good to have since holomorphic functions admit a local series reprsentation.

Definition 5.4 (Complex Analytic)

Given open $U \subset \mathbb{C}$, a function $f : U \to \mathbb{C}$ is $\mathbb{C}$-analytic if for all $D(p,r) \subset U$, there exists a power series

$$\sum_{n=0}^{\infty} a_n (z-p)^n \tag{173}$$

converging to $f(z)$ for all $z \in D(p,r)$.

Theorem 5.2 (Analytic Functions Are Infinitely Differentiable)

Suppose f is $\mathbb{C}$-analytic. Then,

1. $f \in O(U)$ and f' is $\mathbb{C}$-analytic.
2. Therefore, $f^{(k)}$ exists and is $\mathbb{C}$-analytic. Moreover, if $f(z) = \sum_{n=0}^{\infty} a_n (z-p)^n$ in $D(p,r)$, then

$$f^{(k)}(z) = \sum_{n=k}^{\infty} a_n \frac{n!}{(n-k)!}(z-p)^{n-k} \tag{174}$$

   in $D(p,r)$.
3. We have

$$a_n = \frac{f^{(n)}(p)}{n!} \tag{175}$$

Proof. Define $g(z) = \sum_{n=1}^{\infty} n a_n (z-p)^{n-1}$. First, show that this has the same radius of convergence R as f. As an exercise, show that the power series with coefficients $\{a_n\}$ has the same radius of convergence R as the power series with coefficients $\{na_n\}$. The proof is not as obvious because of the $(n-1)$ exponent. That is,

$$R_g^{-1} = \limsup_{n\to\infty} \sqrt[n-1]{n|a_n|} \tag{176}$$

where R_g is the radius of convergence of g. Then,

$$\log R_g = \limsup_{n\to\infty} \frac{1}{n-1} \log |na_n| = \limsup_{n\to\infty} \frac{n}{n-1} \cdot \frac{1}{n} \log |na_n| = \log R_f \tag{177}$$

So being off by 1 doesn't affect the limit. Inductively, it works for every integer. Continuing with the proof, we wish to show that $g(z_0) = f'(z_0)$ for all $z_0 \in D(p,r)$. WLOG, $p = 0$. Fix $\epsilon > 0$. Choose $|z_0| < s < r$. Because of absolute convergence, we know that there exists an $N \geq 2$ s.t.

$$\sum_{n=N+1}^{\infty} n|a_n|s^{n-1} < \epsilon \tag{178}$$

Fix $z = z_0 \in D(0,r)$.

$$\frac{f(z)-f(z_0)}{z-z_0} - g(z_0) = \sum_{n=2}^{\infty} a_n \left(\frac{z^n - z_0^n}{z-z_0} - n z_0^{n-1} \right) \tag{179}$$

where we should write out the polynomial of the quotient. So, let's split the sum

$$= \Big(\sum_{n=2}^{N} + \sum_{n=N+1}^{\infty} \Big) a_n \big(z^{n-1} + z^{n-2} z_0 + \ldots + z z_0^{n-2} + z_0^{n-1} - n z_0^{n-1} \big) \tag{180}$$

Note that the first sum $\sum_{n=2}^{N}$ is finite so it goes to 0 as $z \to z_0$. For the second sum, we know that its absolute value is bounded above by

$$\sum_{n=N+1}^{\infty} \ldots \leq \sum_{n=N+1}^{\infty} 2n |a_n| s^{n-1} < 2\epsilon \tag{181}$$

for $|z| < s$, which implies that

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = g(z_0) \implies f'(z) = \sum_{n=1}^{\infty} n a_n (z-p)^{n-1} \tag{182}$$

(2) follows from induction, and (3) by inspection.

We will do integration later.

6 Jan 20

Previously, we went over the definition.

Definition 6.1 (Holomorphic Function)

Let $U \subset \mathbb{C}$ be open. $f : U \to \mathbb{C}$ is **holomorphic** if

$$f'(p) \coloneqq \lim_{z \to p} \frac{f(z) - f(p)}{z - p} \tag{183}$$

exists for all $p \in U$.

Cauchy's theorem is about integrating holomorphic functions along a curve. Let's talk about integration as a review. Given a C^1 curve $\gamma : [a, b] \to \mathbb{C}$ over compact interval, let's denote $|\gamma| = \{\gamma(t) \in \mathbb{C} \colon a \leq t \leq b\}$ to be the image of γ. Let $f : |\gamma| \to \mathbb{C}$ be continuous, i.e. $f \circ \gamma(t)$ is continuous.

Theorem 6.1 (Mean-Length Inequality)

We have

$$\left| \int_\gamma f(z)\, dz \right| \leq \sup_{z \in |\gamma|} |f(z)| \cdot \text{length}(\gamma) \tag{184}$$

Proof. We have

$$\left| \int_\gamma f(z)\,dz \right| = \left| \int_a^b f(\gamma(t))\gamma'(t)\,dt \right| \tag{185}$$

$$\leq \int_a^b |f(\gamma(t))|\,|\gamma'(t)|\,dt \tag{186}$$

$$\leq \sup_{z\in|\gamma|} |f(z)| \cdot \underbrace{\int_a^b |\gamma'(t)|\,dt}_{\text{length}} \tag{187}$$

Lemma 6.2 (Triangle Integral under Vertex Perturbation)

Let $f : U \to \mathbb{C}$ be continuous, and $T = \triangle abc$ be a closed triangle in U. The integral

$$\int_T f(z)\,dz \tag{188}$$

depends continuously on the vertices. That is, $\forall \epsilon > 0$, $\exists \delta > 0$ s.t.

$$|a-a'|, |b-b'|, |c-c'| < \delta \implies \int_{\partial T} f(z)\,dz - \int_{\partial T'} f(z)\,dz \Bigg| < \epsilon \tag{189}$$

Note that the perturbed triangle must still be in U.

Proof. Decompose this into 3 line integrals.

$$\int_{\partial T} f(z)\,dz = \int_a^b f(z)\,dz + \int_b^c f(z)\,dz + \int_c^a f(z)\,dz \tag{190}$$

So, it suffices to prove for one of the line integrals

$$\int_a^b f(z)\,dz = (b-a)\int_0^1 f\big((1-t)a+tb\big)\,dt \tag{191}$$

where $\gamma(t) = (1-t)a+tb$ for $t \in [0,1]$, with $\gamma'(t) = b-a$. Now, fix open $V \in \mathbb{C}$ s.t. $[a,b] \subset V \subset \overline{V} \subset U$. Note that $\overline{V}$ is compact. Now choose $\delta > 0$ s.t. $|f(z)-f(w)| < \epsilon$ for all $z, w \in V$ s.t. $|z-w| < \delta$. We can do this by uniform continuity of f on $\overline{V}$. Set $\eta(t) = (1-t)a' + tb'$ for $t \in [0,1]$. Then,

$$|\gamma(t) - \eta(t)| \leq (1-t)|a-a'| + t|b-b'| < \delta \tag{192}$$

$$|\gamma'(t) - \eta'(t)| \leq |(b-a)-(b'-a')| < 2\delta \tag{193}$$

And we're basically done, since

$$\left| \int_a^b f(z)\,dz - \int_{a'}^{b'} f(z)\,dz \right| = \left| (b-a)\int_0^1 f\big(\underbrace{(1-t)a+bt}_{z}\big)\,dt - (b'-a')\int_0^1 f\big(\underbrace{(1-t)a'+tb'}_{w}\big)\,dtbigg| \tag{194}$$

...Incomplete proof.

Definition 6.2 (Primitive)

A holomorphic function $F \in O(U)$ is a **primitive** of a continuous $f : U \to \mathbb{C}$ if $F'(z) = f(z)$.

Theorem 6.3 (Existence of a Primitive via Closed Curves)

A continuous function $f : U \to \mathbb{C}$ has a primitive iff

$$\int_\gamma f(z)\,dz = 0 \tag{195}$$

for all closed[a] convex $\gamma : [0,1] \to U$.

[a] in the path sense where $\gamma(0) = \gamma(1)$, not the topological sense

Proof. We prove bidirectionally.

1. ($\to$). Assume that there exists $F \in O(U)$ s.t. $F' = f$. Then,

$$\begin{aligned}
\int_\gamma f(z)\,dz &= \int_0^1 (f \circ \gamma)(t) \cdot \gamma'(t)\,dt && (196)\\
&= \int_0^1 F'(\gamma(t))\,\gamma'(t)\,dt && \text{(substitute)}\\
&= \int_0^1 \frac{d}{dt}(F \circ \gamma)(t)\,dt && \text{(chain rule)}\\
&= F(\gamma(1)) - F(\gamma(0)) && \text{(fund. thm of calc)}\\
&= 0 && \text{(closed)}
\end{aligned}$$

2. ($\leftarrow$). Assume $\int_\gamma f(z)\,dz = 0$. For all closed $\gamma : [0,1] \to U$, WLOG U is connected, and so U is path connected. Fix $p \in U$. Let z be any point in U. Let $\rho, \eta : [0,1] \to U$ be curves s.t.

$$\rho(0) = \eta(0) = p, \qquad \rho(1) = \eta(1) = z \tag{197}$$

we claim that

$$\int_\eta f(w)\,dw = \int_\rho f(w)\,dw \tag{198}$$

i.e. $\int_p^z f(w)\,dw$ doesn't depend on the choice of path from p to z in U. This notation is reserved for such integrals. Consequently, $\int_p^z f(w)\,dw$ is well-defined. To prove this, let's define the reverse path $\rho^- : [0,1] \to U$ as $\rho^-(t) = \rho(1-t)$. Define the closed curve

$$\rho^- * \eta = \gamma : [0,1] \to U, \quad \gamma(t) = \begin{cases} \eta(2t) & \text{if } 0 \le t \le \frac{1}{2} \\ \rho(2t-1) & \text{if } \frac{1}{2} < t \le 1 \end{cases} \tag{199}$$

Then by the hypothesis,

$$\int_\gamma f(w)\,dw = 0 \iff \int_\eta f(w)\,dw = \int_\rho f(w)\,dw \tag{200}$$

Now we claim that $F'(z) = f(z)$. Fix some $z_0 \in U$ and $D(z_0, r) \subset U$ with $r > 0$. Then,

$$F(z) - F(z_0) = \int_{[z_0,z]*\rho} f(w)\,dw - \int_\rho f(w)\,dw = \int_{[z_0,w]} f(w)\,dw \tag{201}$$

$$\implies \frac{F(z) - F(z_0)}{z - z_0} = \frac{1}{z - z_0}\int_{[z_0,z]} f(w)\,dw \tag{202}$$

$$\implies \frac{F(z) - F(z_0)}{z - z_0} - f(z_0) = \frac{1}{z - z_0}\int_{[z_0,z]} f(w) - f(z_0)\,dw \tag{203}$$

Given $\epsilon > 0$, there exists $0 < \delta < r$ s.t. $|f(w) - f(z_0)| < \epsilon$ for all $|w - z_0| < \delta$, which implies

$$\left|\frac{F(z) - F(z_0)}{z - z_0} - f(z)\right| \leq \epsilon \tag{204}$$

for all $|z - z_0| < \delta$.

Example 6.1 (Primitive of z^n for $n \neq -1$)

If $n \neq -1$, then $f(z) = z^n$ has a primitive $F(z) = \frac{z^{n+1}}{n+1}$, which implies

$$\int_\gamma z^n\,dz = 0 \tag{205}$$

for all closed curves $\gamma : [0, 1] \to \mathbb{C}$.

Example 6.2 (The Integral of $1/z$ around a Circle)

If $n = -1$, then this is interesting. Let $\gamma(t) = re^{2\pi it}$ for $t \in [0, 1]$ with $r > 0$ any positive number. This is a parameterization of a circle. Then,

$$\int_\gamma \frac{1}{z}\,dz = \int_0^1 \frac{2\pi i r e^{2\pi it}}{re^{2\pi it}}\,dt = 2\pi i \int_0^1 dt = 2\pi i \tag{206}$$

This is a sneak peek of residue theorem, which implies that $\frac{1}{z}$ is *not* holomorphic in any punctured neighborhood $0 \in \mathbb{C}$. Note that the integral is not depending on radius, so it is "picking up" topological data. $\frac{1}{z}$ also doesn't have primitive in the punctured neighborhood.

Theorem 6.4 (Goursat's Theorem on a Disk)

Suppose $D \subset \mathbb{C}$ is a disk with $f : D \to \mathbb{C}$ continuous. Suppose

$$\int_{\partial T} f(z)\,dz = 0 \tag{207}$$

for all closed triangles $T \subset D$. Then, f has a primitive.

Proof. Define $F(z) \coloneqq \int_{[p,z]} f(w)\,dw$. Then,

$$\int_{\partial T} f(w)\,dw = 0 \implies F(z) - F(z_0) = \int_{[z_0,z]} f(w)\,dw \tag{208}$$

As an exercise, adopt the argument in the proof of the previous theorem (claim 2 specifically) to show that $F'(z_0) = f(z_0)$.

Next, we'll prove Cauchy's theorem and compare $\mathbb{C}$-analytic vs holomorphic.